CAESAR CIPHER:

```
public class JavaApplication {
    public static void main(String[] args) {
        String s = "MEET AT GATE";
        char[] str = s.toCharArray();
        for (int i = 0; i < str.length; i++) {
            char ch = str[i];
            if (Character.isLetter(ch)) {
                char base = Character.isUpperCase(ch) ? 'A' : 'a';
                str[i] = (char) ((ch - base + 3) % 26 + base);
            }
        }
        System.out.println(new String(str));
    }
}
```

OUTPUT:

PHHW DW JDWH

VIGENERE CIPHER:

```
public class VigenereCipher {
  public static void main(String[] args) {
    String plaintext = "MEET AT GATE";
    String keyword = "KEY";

    StringBuilder result = new StringBuilder();
    keyword = keyword.toUpperCase();
    plaintext = plaintext.toUpperCase();
    int keyIndex = 0;

    for (char c : plaintext.toCharArray()) {
      if (c >= 'A' && c <= 'Z') {
        char shift = keyword.charAt(keyIndex % keyword.length());
        char encryptedChar = (char) ((c + shift - 2 * 'A') % 26 + 'A');
        result.append(encryptedChar);
        keyIndex++;
      } else {
        result.append(c);
      }
    }

    System.out.println("Ciphertext: " + result.toString());
  }
}
```

OUTPUT:

Ciphertext: WICD ER QERO

HILL CIPHER:

```
import java.util.*;
class HillCipher {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter text: ");
        String text = sc.nextLine().toUpperCase().replace(" ", "");
        if (text.length() % 2 != 0)
            text += "X";
        String result = "";

        for (int i = 0; i < text.length(); i += 2) {
            int a = text.charAt(i) - 'A';
            int b = text.charAt(i + 1) - 'A';
            int x = (3 * a + 3 * b) % 26;
            int y = (2 * a + 5 * b) % 26;
            result += (char)(x + 'A');
            result += (char)(y + 'A');
        }

        System.out.println("Encrypted text: " + result);
    }
}
```

OUTPUT:

```
Enter text: MEET AT GATE
Encrypted text: WSRZFRSMRG
```

PLAYFAIR CIPHER:

```
import java.util.*;

class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        char[][] a = {
            {'P','L','A','Y','F'},
            {'I','R','B','C','D'},
            {'E','G','H','K','M'},
            {'N','O','Q','S','T'},
            {'U','V','W','X','Z'}
        };
        System.out.print("Enter text:");

        String s = sc.nextLine().toUpperCase().replace(" ", "");
        System.out.println("Key: PLAY FAIR");
        String ans = "";

        for (int k = 0; k < s.length(); k += 2) {

            int p = 0, q = 0;

            for (int i = 0; i < 25; i++) {
                if (a[i/5][i%5] == s.charAt(k))
                    p = i;

                if (a[i/5][i%5] == s.charAt(k+1))
```

```
                q = i;
            }

            int r1 = p/5, c1 = p%5;
            int r2 = q/5, c2 = q%5;

            if (r1 == r2) {
                ans += a[r1][(c1+1)%5];
                ans += a[r2][(c2+1)%5];
            }
            else if (c1 == c2) {
                ans += a[(r1+1)%5][c1];
                ans += a[(r2+1)%5][c2];
            }
            else {
                ans += a[r1][c2];
                ans += a[r2][c1];
            }
        }

        System.out.println("Encrypted:"+ans);
    }
}
```

OUTPUT:

Enter text: MEET AT GATE

Key: PLAY FAIR

Encrypted: EGMNFQHLNM